PROM–ANN, HPROM–ANN, and MAW–ECM

Technical formulation and implementation handoff from the finite-deformation RVE workflow

Project technical note

July 24, 2026

Abstract

This document is the implementation-level description of the RVE workflow in `RVE_homogenization_NeoHookean_using_Kratos`. It explains the complete chain from macro-strain trajectories and full-order snapshots to the structured PROM–ANN coordinates, the intrusive HPROM–ANN correction, the direct D-HPROM–ANN variant, and state-dependent MAW–ECM hyper-reduction. The notation follows the WCCM 2026 slides: the primary coordinates are $\mathbf{q}$, the primary and secondary bases are $\mathbf{V}$ and $\overline{\mathbf{V}}$, and the ANN predicts $\overline{\mathbf{q}}$.

The note is intentionally self-contained. A future implementation for `burgers1d-rom-workbench` should read this file together with the source files listed in Section 9, then port the algorithmic roles rather than copying RVE-specific details such as finite-deformation lifting or stress homogenization.

Contents

1 Purpose, scope, and current RVE configuration

The objective is to approximate the nonlinear solution map of a finite- deformation representative volume element (RVE), while preserving an intrusive equilibrium correction when desired and evaluating both equilibrium and homogenized quantities from a very small set of elements.

The current RVE instance has the following numerical configuration. These numbers document the present experiment; they are *not* assumptions of the method.

Quantity	Current value
Total displacement degrees of freedom	4244
Free displacement degrees of freedom	$N_f = 3924$
Strong Dirichlet degrees of freedom	320
Training snapshots	$N_s = 109460$
Truncated POD dimension	$n_{\text{tot}} = 39$
Primary and secondary dimensions	$n = 3$, $\bar{n} = 36$
Full RVE elements	$N_e = 990$
Final MAW residual support	$\lvert\mathcal{Z}_{\text{res}}\rvert = 10$
Final MAW strain and stress supports	$\lvert\mathcal{Z}_{\varepsilon}\rvert = \lvert\mathcal{Z}_{\sigma}\rvert = 10$
Union of the three final supports	29 elements

There are two online models.

HPROM–ANN The affine map supplies an initial $\mathbf{q}^0$, then a reduced Galerkin/Newton correction solves equilibrium on the ANN-defined nonlinear manifold. MAW–ECM accelerates the residual and the homogenization.

D-HPROM–ANN The direct approximation sets $\mathbf{q} = \mathbf{q}^0$, decodes the state once, and applies the hyper-reduced homogenization. It skips the residual evaluation and all reduced Newton iterations.

2 Full-order RVE problem

2.1 Macro input and affine boundary lifting

For each loading state, the physical macro input is either a deformation gradient $\mathbf{F}$ or, equivalently for the present workflow, its Green– Lagrange strain

$$\mathbf{E} = \frac{1}{2}\left(\mathbf{F}^{\mathsf{T}}\mathbf{F} - \mathbf{I}\right), \qquad \mu = \begin{bmatrix} E_{xx} & E_{yy} & G_{xy} \end{bmatrix}^{\mathsf{T}} \in \mathbb{R}^3. \tag{1}$$

The code reconstructs the chosen two-dimensional branch of $\mathbf{F}$ from $\mathbf{E}$. Let $\mathbf{X}$ denote a reference material point and $\mathbf{X}_c$ the RVE reference-domain center. The imposed affine displacement is

$$\mathbf{u}_{\text{aff}}(\mathbf{X};\mu) = (\mathbf{F}(\mu) - \mathbf{I})\,(\mathbf{X} - \mathbf{X}_c). \tag{2}$$

The total displacement is decomposed as

$$\mathbf{u}(\mu) = \mathbf{u}_{\text{aff}}(\mu) + \mathbf{s}(\mu), \tag{3}$$

where the fluctuation $\mathbf{s}$ is zero on the strongly prescribed Dirichlet degrees of freedom. All reduced bases and coordinates below act on the N_f free entries of $\mathbf{s}$.

At element level, let $\mathbf{r}_e(\mathbf{u};\mu)$ and $\mathbf{K}_e(\mathbf{u};\mu)$ denote the element residual and its tangent. The full internal equilibrium residual and tangent are assembled as

$$\mathbf{r}(\mathbf{u};\mu) = \sum_{e=1}^{N_e} \mathbf{r}_e(\mathbf{u};\mu), \qquad \mathbf{K}(\mathbf{u};\mu) = \sum_{e=1}^{N_e} \mathbf{K}_e(\mathbf{u};\mu). \tag{4}$$

The full-order model (FOM) solves $\mathbf{r} = \mathbf{0}$, with the sign convention used by the underlying finite-element assembly. All reduced equations must retain that convention consistently.

2.2 Homogenized strain and stress

Let A_e be the reference measure of element e, $A_0 = \sum_e A_e$ the reference RVE measure, and $\langle\cdot\rangle_e$ the element Gauss-point average. The code returns the same strain and stress measures that are stored by the finite-element constitutive law. Their full-order homogenizations have the generic form

$$\overline{\mathbf{E}} = \frac{1}{A_0}\sum_{e=1}^{N_e} A_e \left\langle \mathbf{E}_e \right\rangle_e, \qquad \overline{\mathbf{S}} = \frac{1}{A_0}\sum_{e=1}^{N_e} A_e \left\langle \mathbf{S}_e \right\rangle_e. \tag{5}$$

The symbol $\mathbf{S}$ in Eq. (5) is a stress tensor; the snapshot matrix later in the document is written in bold as $\mathbf{S}$ only within the POD section. Context and dimensions distinguish the two.

3 Training trajectories and FOM data

3.1 Macro-strain domain

The current training domain is a box in the three physical macro-strain components. With a prescribed positive and negative extent in each direction, it can be written as

$$\mathcal{D}_\mu = [E_{xx}^-, E_{xx}^+] \times [E_{yy}^-, E_{yy}^+] \times [G_{xy}^-, G_{xy}^+]. \tag{6}$$

For the WCCM data, `stage0_training_trajectory.py` uses $E_{\max} = 2$ and relative bounds

$$(1, 0.05, 1, 0.05, 0.05, 0.05), \tag{7}$$

giving the asymmetric box used in the trajectory figures.

3.2 Serpentine paths

The loading set is deliberately a collection of continuous paths rather than an unordered cloud. Five shear layers are used,

$$G_{xy} \in \{0, +G_1, +G_2, -G_1, -G_2\}, \tag{8}$$

and each layer contains two scans starting from the origin: an upper scan and a lower scan. Within each fixed-shear plane, the trajectory visits a sequence of E_{yy} rows and alternates the direction of its E_{xx} sweep. Consequently, the default construction contains ten continuous serpentine trajectories. This gives coverage of the macro domain while avoiding unrelated jumps between FOM states.

For a path with waypoints $\{\mu_\ell\}_{\ell=0}^{L}$, the actual FOM path is piecewise linear,

$$\mu(\lambda) = (1-\lambda)\mu_\ell + \lambda\mu_{\ell+1}, \qquad \lambda \in [0,1]. \tag{9}$$

The number of increments on a segment is proportional to its scaled length. In the present script the visualization and increment metric use

$$\widehat{\mu} = [E_{xx}, E_{yy}, G_{xy}/2]^\mathsf{T}, \qquad n_\ell = \max\left(1, \left\lceil n_{\mathrm{ref}} \frac{\|\widehat{\mu}_{\ell+1} - \widehat{\mu}_\ell\|}{a_{\mathrm{ref}}} \right\rceil\right). \tag{10}$$

The scaling is an RVE trajectory-design choice, not part of PROM–ANN. Its purpose is to keep nonlinear increments manageable and maintain a comparable spatial resolution on every path.

3.3 Snapshot collection

The FOM is run on every trajectory. At each increment j, it stores

$$\left\{ \mathbf{u}^{(j)},\ \mu^{(j)},\ \overline{\mathbf{E}}^{(j)},\ \overline{\mathbf{S}}^{(j)} \right\}. \tag{11}$$

The affine-free free-DOF snapshot is

$$\mathbf{s}^{(j)} = \left(\mathbf{u}^{(j)} - \mathbf{u}_{\text{aff}}(\mu^{(j)}) \right)\Big|_{\text{free}} \in \mathbb{R}^{N_f}. \tag{12}$$

The exact vanishing of the Dirichlet entries of this quantity is checked in Stage 2. It is essential: otherwise the reduced coordinates would mix a known boundary lifting with the unknown fluctuation.

4 POD and the structured coordinate system

4.1 Truncated POD representation

Collect the snapshots as columns:

$$\mathbf{S} = \begin{bmatrix} \mathbf{s}^{(1)} & \cdots & \mathbf{s}^{(N_s)} \end{bmatrix} \in \mathbb{R}^{N_f \times N_s}. \tag{13}$$

The POD of $\mathbf{S}$ gives the truncated orthonormal basis $\mathbf{V}_{\text{tot}} \in \mathbb{R}^{N_f \times n_{\text{tot}}}$. In the present RVE, $n_{\text{tot}} = 39$ captures $1 - 9.01 \times 10^{-11}$ of the snapshot energy. Define the standard POD coordinate matrix

$$\mathbf{Q}_{\text{tot}} = \mathbf{V}_{\text{tot}}^{\mathsf{T}} \mathbf{S} \in \mathbb{R}^{n_{\text{tot}} \times N_s}, \qquad \mathbf{S} \approx \mathbf{V}_{\text{tot}} \mathbf{Q}_{\text{tot}}. \tag{14}$$

The first three standard POD coordinates are excellent for minimizing a linear reconstruction error. They need not be the best three coordinates for regression from μ: a nonlinear loading manifold can appear curved, coupled, or folded when projected onto those first modes. The following construction therefore uses *all* retained POD coordinates to identify a three-dimensional subspace aligned with macro loading.

4.2 Linear alignment with macro strain

Collect the same macro inputs as columns,

$$\mathbf{M} = \begin{bmatrix} \mu^{(1)} & \cdots & \mu^{(N_s)} \end{bmatrix} \in \mathbb{R}^{3 \times N_s}. \tag{15}$$

The least-squares alignment matrix is

$$\mathbf{T} = \underset{\widehat{\mathbf{T}} \in \mathbb{R}^{3 \times n_{\text{tot}}}}{\arg\min} \left\| \widehat{\mathbf{T}} \mathbf{Q}_{\text{tot}} - \mathbf{M} \right\|_{\text{F}}^{2}. \tag{16}$$

Thus $\mathbf{T}\mathbf{Q}_{\text{tot}}$ is the best linear prediction of the three macro-strain components available from the entire retained POD coordinate set. The implementation stores the row-sample equivalent of Eq. (16): $\mathbf{Q}_{\text{tot}}^{\mathsf{T}} \mathbf{T}^{\mathsf{T}} \approx \mathbf{M}^{\mathsf{T}}$.

Map the aligned directions to physical free-DOF space and factor them by a compact singular value decomposition,

$$\mathbf{V}_{\text{tot}} \mathbf{T}^{\mathsf{T}} = \mathbf{U} \boldsymbol{\Sigma} \mathbf{R}^{\mathsf{T}} = \mathbf{V}\mathbf{A}, \qquad \mathbf{V} = \mathbf{U} \in \mathbb{R}^{N_f \times 3}, \qquad \mathbf{A} = \boldsymbol{\Sigma} \mathbf{R}^{\mathsf{T}} \in \mathbb{R}^{3 \times 3}. \tag{17}$$

The matrix $\mathbf{V}$ is orthonormal. The non-orthogonal scaling and rotation required by the macro-aligned directions are retained in $\mathbf{A}$. This is why the state decoder contains $\mathbf{V}\mathbf{A}\mathbf{q}$, not merely $\mathbf{V}\mathbf{q}$.

4.3 Primary and secondary coordinates

For an affine-free state $\mathbf{s}$, define the structured primary coordinates by

$$\mathbf{q} = \mathbf{A}^{-1}\mathbf{V}^{\mathsf{T}}\mathbf{s} \in \mathbb{R}^3. \tag{18}$$

For the whole data set this gives

$$\mathbf{V}^{\mathsf{T}}\mathbf{S} = \mathbf{A}\mathbf{Q}, \qquad \mathbf{Q} = \mathbf{A}^{-1}\mathbf{V}^{\mathsf{T}}\mathbf{S} \in \mathbb{R}^{3\times N_s}. \tag{19}$$

The remaining directions are an orthonormal complement in the retained POD space. In the code, if $\mathbf{C}_m = \mathbf{V}_{\mathrm{tot}}^{\mathsf{T}}\mathbf{V}$, then $\mathbf{C}_s$ spans the complement of $\mathbf{C}_m$, and

$$\overline{\mathbf{V}} = \mathbf{V}_{\mathrm{tot}}\mathbf{C}_s, \qquad \overline{\mathbf{Q}} = \overline{\mathbf{V}}^{\mathsf{T}}\mathbf{S} \in \mathbb{R}^{36\times N_s}. \tag{20}$$

The truncated-POD reconstruction is therefore equivalently

$$\mathbf{S} \approx \mathbf{V}\mathbf{A}\mathbf{Q} + \overline{\mathbf{V}\mathbf{Q}}. \tag{21}$$

Important distinction. The coordinates $\mathbf{q}$ are *not* the first three raw standard POD coordinates. They are the coordinates of a three-dimensional subspace identified from all 39 retained modes through Eq. (16) and then factored through Eq. (17). This is the precise meaning of "structured coordinates better suited to regression."

5 PROM–ANN state approximation

5.1 Affine primary-coordinate initialization

The map from macro strain to the structured primary coordinate is very nearly affine in the current data. Define the row-sample design matrix

$$\mathbf{M}_{\mathrm{aff}} = \begin{bmatrix}\mathbf{M}^{\mathsf{T}} & \mathbf{1}\end{bmatrix} \in \mathbb{R}^{N_s\times 4}. \tag{22}$$

The coefficient matrix $\mathbf{B}_{\mathrm{aff}} \in \mathbb{R}^{4\times 3}$ is obtained by

$$\mathbf{B}_{\mathrm{aff}} = \underset{\mathbf{B}\in\mathbb{R}^{4\times 3}}{\arg\min} \left\|\mathbf{M}_{\mathrm{aff}}\mathbf{B} - \mathbf{Q}^{\mathsf{T}}\right\|_{\mathrm{F}}^{2}. \tag{23}$$

For one macro state, the affine prediction is

$$\left(\mathbf{q}^0(\mu)\right)^{\mathsf{T}} = \begin{bmatrix}\mu^{\mathsf{T}} & 1\end{bmatrix}\mathbf{B}_{\mathrm{aff}}. \tag{24}$$

In the current RVE, the relative fit error of this map is 6.88×10^{-4}. This makes $\mathbf{q}^0$ an excellent initialization for an intrusive solve and a meaningful direct prediction.

The role of $\mathbf{A}$ deserves a precise statement. It influences $\mathbf{q}$ through Eq. (18), so it influences the data used to fit $\mathbf{B}_{\mathrm{aff}}$. However, $\mathbf{A}$ does *not* appear explicitly in the least-squares problem Eq. (23); that problem directly fits $\mu \mapsto \mathbf{q}$.

5.2 ANN closure of secondary coordinates

The closure network learns the discarded coordinates as a function of the three structured coordinates,

$$\overline{\mathbf{q}} \approx \mathcal{N}_\theta(\mathbf{q}), \qquad \mathbf{q} \in \mathbb{R}^3, \qquad \overline{\mathbf{q}} \in \mathbb{R}^{36}. \tag{25}$$

It is trained from the pairs $\{\mathbf{q}^{(j)}, \overline{\mathbf{q}}^{(j)}\}$. The production solver uses a closure with its constant and linear terms removed:

$$\mathcal{N}_\theta^\circ(\mathbf{q}) = \mathcal{N}_\theta(\mathbf{q}) - \mathcal{N}_\theta(\mathbf{0}) - \left.\frac{\partial \mathcal{N}_\theta}{\partial \mathbf{q}}\right|_{\mathbf{0}} \mathbf{q}. \tag{26}$$

Hence $\mathcal{N}_\theta^\circ(\mathbf{0}) = \mathbf{0}$ and $\partial \mathcal{N}_\theta^\circ / \partial \mathbf{q}(\mathbf{0}) = \mathbf{0}$. For concise notation, the superscript $\circ$ is omitted below and $\mathcal{N}_\theta$ denotes the corrected closure.

The nonlinear PROM–ANN decoder is

$$\tilde{\mathbf{u}}(\mathbf{q}, \mu) = \mathbf{u}_{\text{aff}}(\mu) + \mathbf{V}\mathbf{A}\mathbf{q} + \overline{\mathbf{V}}\mathcal{N}_\theta(\mathbf{q}). \tag{27}$$

Its tangent with respect to the primary coordinates is the nonlinear-manifold test basis

$$\mathbf{W}(\mathbf{q}) = \frac{\partial \tilde{\mathbf{u}}}{\partial \mathbf{q}} = \mathbf{V}\mathbf{A} + \overline{\mathbf{V}}\frac{\partial \mathcal{N}_\theta}{\partial \mathbf{q}}(\mathbf{q}) \in \mathbb{R}^{N_f \times 3}. \tag{28}$$

The restriction of $\mathbf{W}$ to element e is denoted by $\mathbf{W}_e$.

6 Intrusive PROM–ANN and HPROM–ANN

6.1 PROM–ANN Galerkin correction

The intrusive reduced equilibrium equation is a Galerkin projection using the tangent of the nonlinear decoder:

$$\mathbf{g}(\mathbf{q}; \mu) = \mathbf{W}(\mathbf{q})^{\mathsf{T}} \mathbf{r}(\tilde{\mathbf{u}}(\mathbf{q}, \mu); \mu) = \mathbf{0}. \tag{29}$$

The online solve initializes from $\mathbf{q}^0(\mu)$ in Eq. (24), then applies reduced Newton iterations,

$$\mathbf{K}_{\text{red}}(\mathbf{q}^k)\Delta\mathbf{q}^k = -\mathbf{g}(\mathbf{q}^k; \mu), \qquad \mathbf{q}^{k+1} = \mathbf{q}^k + \Delta\mathbf{q}^k. \tag{30}$$

Ignoring manifold curvature gives the standard Gauss–Newton-style reduced tangent

$$\mathbf{K}_{\text{GN}} = \mathbf{W}^{\mathsf{T}}\mathbf{K}\mathbf{W}. \tag{31}$$

The exact derivative of Eq. (29) also contains the curvature term $\mathrm{D}\mathbf{W}^{\mathsf{T}}[\mathbf{r}]$. The code has an option to include that term. This distinction is separate from MAW–ECM.

6.2 Elementwise hyper-reduction

Directly forming Eq. (29) still visits all N_e elements. For a fixed empirical cubature rule with support $\mathcal{Z}_{\text{res}}$ and weights ξ_e^{res}, the hyper-reduced Galerkin residual is

$$\mathbf{g}_{\text{HR}}(\mathbf{q}; \mu) \approx \sum_{e \in \mathcal{Z}_{\text{res}}} \xi_e^{\text{res}}\, \mathbf{W}_e(\mathbf{q})^{\mathsf{T}} \mathbf{r}_e(\tilde{\mathbf{u}}(\mathbf{q}, \mu); \mu)\,. \tag{32}$$

Classical ECM selects a small positive support and weights offline, after a randomized SVD compression of sampled elementwise operators. Stage 9 builds the residual data block for sampled state j as

$$\mathbf{Q}_j^{\text{res}} = \begin{bmatrix} \mathbf{W}_1^{\mathsf{T}}\mathbf{r}_1 & \cdots & \mathbf{W}_{N_e}^{\mathsf{T}}\mathbf{r}_{N_e} \end{bmatrix} \in \mathbb{R}^{3 \times N_e}, \qquad \mathbf{b}_j^{\text{res}} = \mathbf{Q}_j^{\text{res}}\mathbf{1}. \tag{33}$$

The column associated with element e is exactly its contribution to the Galerkin residual. Fixed ECM finds a support and weights that reproduce the sampled blocks to the requested tolerance.

6.3 D-HPROM–ANN

The direct variant is not an equilibrium solve. It uses

$$\mathbf{q} = \mathbf{q}^0(\mu), \qquad \tilde{\mathbf{u}}_{\mathrm{D}} = \tilde{\mathbf{u}}(\mathbf{q}^0(\mu), \mu), \tag{34}$$

then evaluates the homogenized quantities with the hyper-reduced output rule in Section 8.2. It does not evaluate Eq. (32) and does not perform the correction Eq. (30). Therefore its quality relies on the affine initialization and ANN decoder being sufficiently accurate over the deployment domain.

7 MAW–ECM: state-dependent cubature weights

7.1 Core idea

MAW–ECM keeps a *common sparse element support* but allows its weights to vary smoothly with the state. For any target τ (residual, strain, or stress), let $\mathcal{Z}_\tau$ be its selected support and let $\xi^\tau(\mathbf{z})$ contain the support weights. The generic hyper-reduced operator is

$$\sum_{e=1}^{N_e} \mathbf{a}_e^\tau \approx \sum_{e\in\mathcal{Z}_\tau} \xi_e^\tau(\mathbf{z})\mathbf{a}_e^\tau. \tag{35}$$

The support is fixed after offline pruning; only $\xi^\tau(\mathbf{z})$ changes online. This is important for a reduced mesh implementation because the mesh connectivity can stay fixed.

7.2 Local MAW constraints and common-support pruning

Start from a feasible classical ECM candidate support $\mathcal{Z}_{\tau,\mathrm{ini}}$ with weights ξ^τ_{ini}. At each training state $\mathbf{z}_j$, build a local linear constraint

$$\mathbf{A}_j^\tau \xi_j^\tau = \mathbf{b}_j^\tau, \qquad \xi_j^\tau \geq \mathbf{0}, \qquad \mathbf{1}^{\mathsf{T}}\xi_j^\tau = \alpha_\tau. \tag{36}$$

The last equality is used in the present final model with $\alpha_\tau = 990$, the number of full RVE elements. It preserves the reference integration measure and makes the weights nonnegative and comparable across states.

For residual MAW,

$$\mathbf{A}_j^{\mathrm{res}} = \mathbf{Q}_j^{\mathrm{res}}(:, \mathcal{Z}_{\mathrm{res,ini}}). \tag{37}$$

For strain and stress MAW, the elementwise homogenization data are

$$\mathbf{C}_j^\varepsilon(:, e) = A_e \left\langle \mathbf{E}_e \right\rangle_e, \qquad \mathbf{C}_j^\sigma(:, e) = A_e \left\langle \mathbf{S}_e \right\rangle_e, \tag{38}$$

and the corresponding $\mathbf{A}_j^\tau$ is the restriction of the required columns to the candidate support. In a full-target variant, $\mathbf{b}_j^\tau$ is the sum over all elements. In the final RVE MAW run, `rhs_mode=anchor`: the right hand side is

$$\mathbf{b}_j^\tau = \mathbf{A}_j^\tau \xi_{\mathrm{ini}}^\tau. \tag{39}$$

This preserves the classical ECM rule locally while MAW pruning seeks a smaller common support and a smooth state-dependent weight field.

The pruning routine eliminates candidate elements while maintaining the feasibility of Eq. (36) over all sampled states. Its optional second phase introduces state-space smoothness, schematically

$$\min_{\{\xi_j\}} \sum_{(i,j)\in\mathcal{E}_{\mathrm{graph}}} \|\xi_i - \xi_j\|^2 \quad \text{subject to the local constraints.} \tag{40}$$

The exact pruning procedure is an active-set elimination algorithm; it returns the common support and the feasible training matrix

$$\boldsymbol{\Xi}_\tau = \left[\xi^\tau(\mathbf{z}_1) \quad \cdots \quad \xi^\tau(\mathbf{z}_{N_\tau})\right]. \tag{41}$$

7.3 Weight regressors

The support weights must be evaluated at new online states. The final RVE uses two regressor families.

Residual MAW–RBF. The residual support weights depend on the structured coordinates, $\mathbf{z}_{\text{res}} = \mathbf{q}$. A multi-output Gaussian RBF has the form

$$\widehat{\xi}^{\text{res}}(\mathbf{q}) = \mathbf{D}\left[\mathbf{\Phi}(\mathbf{q})\mathbf{A}_{\text{RBF}} + \mathbf{p}(\mathbf{q})\mathbf{B}_{\text{RBF}}\right]^{\mathsf{T}}, \qquad \Phi_{\ell}(\mathbf{q}) = \exp\left[-\frac{1}{2}\sum_{a}\left(\frac{q_a - c_{\ell a}}{\ell_a}\right)^2\right]. \tag{42}$$

Here $\mathbf{D}$ restores output scaling and $\mathbf{p}$ is an optional constant or affine polynomial block. The online implementation clips to nonnegative values and renormalizes the weight sum to α_{res}. It also evaluates $\partial\xi^{\text{res}}/\partial\mathbf{q}$ analytically.

Homogenization MAW–ANN. The final strain and stress support weights depend on macro input,

$$\mathbf{z}_{\varepsilon} = \mathbf{z}_{\sigma} = \mu. \tag{43}$$

Their ANN output is converted to positive, sum-preserving weights by

$$\xi^{\tau}(\mu) = \alpha_{\tau}\ \text{softmax}\left(\mathcal{W}_{\omega}^{\tau}(\mu)\right), \qquad \tau \in \{\varepsilon, \sigma\}. \tag{44}$$

This gives nonnegative weights with exactly the desired total by construction.

Current final MAW model. The saved file `stage_12_hprom_ann_mawecm_res_eps_sig_phase1to40_phase2to10_sum990_ann/ecm_weights_all.npz` uses a three-dimensional $\mathbf{q}$, an RBF residual-weight map, ANN strain and stress weight maps driven by μ, and ten elements per target.

8 HPROM–ANN with MAW–ECM

8.1 Dynamic residual and consistent tangent

With state-dependent residual weights, the online projected residual becomes

$$\mathbf{g}_{\text{MAW}}(\mathbf{q};\mu) = \sum_{e\in\mathcal{Z}_{\text{res}}} \xi_e^{\text{res}}(\mathbf{q})\mathbf{W}_e(\mathbf{q})^{\mathsf{T}}\mathbf{r}_e(\tilde{\mathbf{u}}(\mathbf{q},\mu);\mu)\,. \tag{45}$$

Let $\mathbf{a}_e = \mathbf{W}_e^{\mathsf{T}}\mathbf{r}_e$. Its exact directional derivative has three contributions,

$$\mathrm{D}\mathbf{g}_{\text{MAW}} = \sum_{e\in\mathcal{Z}_{\text{res}}}\left[\xi_e^{\text{res}}\,(\mathrm{D}\mathbf{W}_e^{\mathsf{T}})[\mathbf{r}_e] + \xi_e^{\text{res}}\,\mathbf{W}_e^{\mathsf{T}}\mathbf{K}_e\mathbf{W}_e + \mathbf{a}_e\,(\nabla_{\mathbf{q}}\xi_e^{\text{res}})^{\mathsf{T}}\right]. \tag{46}$$

The first term is the nonlinear-manifold curvature, the second is the usual Galerkin stiffness term, and the third is the MAW weight-tangent term. If the weights are fixed, the last term vanishes. If a Gauss–Newton manifold tangent is used, the first term is neglected. The production HPROM–ANN solver can include both the manifold curvature and the MAW weight tangent; the state-dependent weight tangent must not be silently omitted when using Eq. (45).

The Newton correction is

$$\mathbf{K}_{\text{MAW}}(\mathbf{q}^k)\Delta\mathbf{q}^k = -\mathbf{g}_{\text{MAW}}(\mathbf{q}^k;\mu), \qquad \mathbf{q}^{k+1} = \mathbf{q}^k + \Delta\mathbf{q}^k. \tag{47}$$

8.2 Hyper-reduced homogenization

After obtaining the accepted state, homogenized strain and stress are approximated from their own selected supports:

$$\overline{\mathbf{E}}_{\mathrm{HR}} \approx \frac{1}{A_0} \sum_{e \in \mathcal{Z}_\varepsilon} \xi_e^\varepsilon(\mu) A_e \left\langle \mathbf{E}_e \right\rangle_e , \tag{48}$$

$$\overline{\mathbf{S}}_{\mathrm{HR}} \approx \frac{1}{A_0} \sum_{e \in \mathcal{Z}_\sigma} \xi_e^\sigma(\mu) A_e \left\langle \mathbf{S}_e \right\rangle_e . \tag{49}$$

The residual, strain, and stress supports may overlap, but they have distinct purposes and should be treated as independent rules. A reduced HROM mesh is formed from their union plus the nodes and conditions required by the strong boundary conditions.

8.3 Online algorithms

HPROM–ANN with MAW–ECM. For each macro increment μ^n:

1. Compute $\mathbf{F}(\mu^n)$ and $\mathbf{u}_{\mathrm{aff}}(\mu^n)$.
2. Form $\mathbf{q}^0(\mu^n)$ from Eq. (24). Continuation from the previous converged $\mathbf{q}$ may be used after the first step.
3. Decode $\tilde{\mathbf{u}}$, evaluate $\mathbf{W}$, and evaluate the residual MAW weights and their derivative.
4. Assemble Eq. (45) and its selected tangent terms, then solve Eq. (47) until the reduced convergence criterion is met.
5. Evaluate Eq. (48)–Eq. (49) with the accepted state and the strain/stress MAW weights.

D-HPROM–ANN. For each macro increment μ^n:

1. Compute $\mathbf{q} = \mathbf{q}^0(\mu^n)$.
2. Decode $\tilde{\mathbf{u}}_{\mathrm{D}}$ through Eq. (27).
3. Evaluate only the hyper-reduced homogenization Eq. (48)–Eq. (49).

This direct path is appropriate only after verifying that its lack of equilibrium correction is acceptable on unseen trajectories.

9 Offline workflow and source map

Stage	Main files	Role
0	`stage0_training_ trajectory.py`	Builds the ten serpentine macro-strain paths and writes `stage_0_trajectories.npz`.
1	`stage1_training_ fom_solver_rve.py`, `fom_solver_rve.py`	Runs the FOM for each path and saves displacement, macro-strain, homogenized strain, and homogenized stress histories.
2	`stage2_pod_rve.py`	Removes $\mathbf{u}_{\mathrm{aff}}$, checks strong Dirichlet consistency, and computes $\mathbf{V}_{\mathrm{tot}}$.

Stage	Main files	Role
7a	`stage7a_prepare_rbf_dataset_ls.py`; wrapper `stage7a_prepare_ann_dataset_ls.py`	Implements Eq. (16)–Eq. (23); saves $\mathbf{V}$, $\overline{\mathbf{V}}$, $\mathbf{A}$, $\mathbf{B}_{\text{aff}}$, and ANN data.
7b	`stage7b_train_ann_manifold.py`	Trains $\mathcal{N}_\theta : \mathbf{q} \mapsto \overline{\mathbf{q}}$.
9a	`stage9_build_ecm_dataset_ann.py`	Builds sampled elementwise residual and homogenization operators.
9b	`stage9_compute_ecm_weights_ann.py`	Builds the classical fixed ECM rule used as the MAW anchor.
10	`stage10_test_hprom_ann_ls.py`, `hprom_ann_solver_rve.py`	Runs and benchmarks PROM–ANN, HPROM–ANN, and D-HPROM–ANN.
12	`stage12b_build_mawecm_model_gpr.py`, `mawecm_pruning.py`, `mawecm_rbf_weights.py`, `mawecm_ann_weights.py`	Builds the common supports, local feasible MAW weights, and online RBF/ANN weight regressors. The builder name is historical; it supports the ANN workflow.

The two most important persisted artifacts for the current case are:

- `stage_7_ann_data_ls/`: structured coordinates, primary/secondary bases, $\mathbf{A}$, $\mathbf{T}$, and the affine initialization fit;
- `stage_12_hprom_ann_mawecm_res_eps_sig_phase1to40_phase2to10_sum990_ann/ecm_weights_all.npz`: all MAW supports, models, HROM mesh metadata, and diagnostic data.

10 Validation logic and error decomposition

The method has several distinct approximation layers. They should be validated separately before interpreting a final stress error.

1. **POD truncation:** compare $\mathbf{s}$ with $\mathbf{V}_{\text{tot}}\mathbf{V}_{\text{tot}}^{\mathsf{T}}\mathbf{s}$.
2. **Coordinate construction:** check $\|\mathbf{T}\mathbf{Q}_{\text{tot}} - \mathbf{M}\|_{\text{F}}$, the invertibility and conditioning of $\mathbf{A}$, and the exact split reconstruction in Eq. (21).
3. **Affine initialization:** evaluate the unseen-trajectory error of $\mathbf{q}^0(\mu)$ from Eq. (24).
4. **ANN closure:** evaluate the error in $\overline{\mathbf{q}}$, in the decoded state, and in the nonlinear tangent $\mathbf{W}$ if it is used intrusively.
5. **Fixed ECM and MAW constraints:** verify every sampled local constraint $\mathbf{A}_j\xi_j = \mathbf{b}_j$, positivity, and the required weight sum. Validate regressed weights away from the fitting nodes.
6. **HPROM correction:** compare the corrected $\mathbf{q}$ and reduced residual convergence against the non-hyper-reduced PROM–ANN.
7. **Output accuracy:** compare FOM, PROM–ANN, HPROM–ANN, and D-HPROM–ANN homogenized stress histories on trajectories excluded from the offline construction.

This order localizes failures. For example, a good ANN state with a poor stress curve points to output hyper-reduction, whereas a poor direct state but good corrected HPROM state points to $\mathbf{q}^0$, not to MAW–ECM.

11 Porting the technology to Burgers 1D

11.1 What transfers unchanged

The reusable architecture is independent of the RVE:

$$\begin{aligned}&\text{training trajectories} \longrightarrow \text{snapshots} \longrightarrow \text{POD} \longrightarrow \text{structured coordinates}\\ &\longrightarrow \text{ANN closure} \longrightarrow \text{Galerkin correction and MAW–ECM.}\end{aligned} \tag{50}$$

For Burgers, replace the RVE state, residual, and homogenized stress by the finite-element or finite-volume Burgers state, time-discrete residual, and the desired output functional. The coordinate construction, $\mathbf{VAq} + \overline{\mathbf{V}}\mathcal{N}_\theta(\mathbf{q})$, and the elementwise MAW residual construction remain the same.

11.2 Transient residual

For example, a backward-Euler semi-discretization of Burgers at time step n has a residual of the generic form

$$\mathbf{r}^n(\mathbf{u}^n; \mathbf{u}^{n-1}, \mu^n) = \frac{1}{\Delta t}\mathbf{M}_h(\mathbf{u}^n - \mathbf{u}^{n-1}) + \mathbf{f}_{\mathrm{conv}}(\mathbf{u}^n) + \mathbf{f}_{\mathrm{diff}}(\mathbf{u}^n; \mu^n) - \mathbf{f}_{\mathrm{ext}}(\mu^n). \tag{51}$$

The PROM–ANN Galerkin residual becomes

$$\mathbf{g}^n(\mathbf{q}^n) = \mathbf{W}(\mathbf{q}^n)^\top \mathbf{r}^n\left(\tilde{\mathbf{u}}(\mathbf{q}^n, \mu^n); \tilde{\mathbf{u}}(\mathbf{q}^{n-1}, \mu^{n-1}), \mu^n\right). \tag{52}$$

The elementwise residual operator used by ECM must include both the spatial terms and the time-discrete mass contribution assigned to each element.

11.3 Critical modeling decision for a transient problem

For the RVE, μ identifies a quasi-static state well enough that the affine map $\mathbf{q}^0(\mu)$ is highly accurate. This is not automatically true for Burgers. Two states can have the same viscosity or forcing parameter but differ because of time history. Before porting Eq. (23), decide what identifies the current state:

1. If the dynamics are deterministic from a known initial condition, include time and the active forcing descriptors in μ^n, then fit $\mathbf{q}^0(\mu^n)$.
2. If history matters beyond the chosen input, use continuation $\mathbf{q}^{0,n} = \mathbf{q}^{n-1}$ as the principal initializer and retain the intrusive reduced solve.
3. If a direct model is needed, augment the ANN input with enough state history or learn a recurrent/time-step map. Do not claim a direct map from an insufficient parameter vector to a history-dependent state.

11.4 Burgers-specific offline plan

1. Define parameter/time trajectories that cover viscosity, forcing, boundary conditions, initial-condition families, and the time interval of interest.
2. Generate FOM snapshots. If inhomogeneous Dirichlet conditions are present, remove an exact lifting as in Eq. (2); otherwise use the state directly.
3. Compute a POD basis and select the primary dimension from the parameter/state dimension that is physically meaningful for the intended latent representation.

4. Fit $\mathbf{T}$ using Eq. (16), factor $\mathbf{V}_{\text{tot}}\mathbf{T}^\top = \mathbf{VA}$, form $\mathbf{q}$, and train the secondary-coordinate closure.

5. Build residual ECM blocks using $\mathbf{W}_e^\top \mathbf{r}_e^n$ at sampled time states. Build a separate output ECM block only if a spatial integral, flux, energy, or other output requires it.

6. Build classical ECM first, then MAW supports and weight regressors. The residual MAW input should normally be $\mathbf{q}$; an output MAW input can be $\mathbf{q}$, μ, or both, provided it identifies the integrand reliably.

7. Validate fixed PROM, fixed HPROM, MAW HPROM, and a direct variant on unseen trajectories, including long-time stability and residual convergence.

12 Implementation checklist for a future coding session

Before implementing the Burgers version, establish the following explicitly.

1. Snapshot orientation: this document uses states as columns in $\mathbf{S}$, whereas several NumPy files store samples as rows. Transpose equations consistently rather than mixing conventions.

2. Lifting: define whether the reduced state is the full field or an affine-free fluctuation. Do not build a basis from mixed boundary data.

3. Coordinate definition: retain the exact sequence $\mathbf{Q}_{\text{tot}}$, $\mathbf{T}$, $\mathbf{VA}$, then $\mathbf{Q}$. Avoid introducing a second, ambiguously named "master" coordinate.

4. Decoder: use $\tilde{\mathbf{u}} = \mathbf{u}_{\text{aff}} + \mathbf{VAq} + \overline{\mathbf{V}}\mathcal{N}_\theta(\mathbf{q})$, and differentiate it to obtain the Galerkin test basis $\mathbf{W}$.

5. Direct versus intrusive path: the direct path is an approximation; the intrusive path enforces a projected residual. Benchmark both instead of assuming that a good closure automatically gives a good direct solver.

6. MAW: supports are fixed, weights are dynamic. Include $\partial\xi/\partial\mathbf{q}$ in the reduced tangent when residual weights depend on $\mathbf{q}$.

7. Outputs: use a separate hyper-reduced rule for every quantity whose elementwise integrand differs materially from the residual.

13 Compact formula reference

$$
\begin{aligned}
\mathbf{E} &= \tfrac{1}{2}(\mathbf{F}^{\mathsf{T}}\mathbf{F} - \mathbf{I}), & \mu &= [E_{xx}, E_{yy}, G_{xy}]^{\mathsf{T}}, && (53)\\
\mathbf{u} &= \mathbf{u}_{\text{aff}} + \mathbf{s}, & \mathbf{Q}_{\text{tot}} &= \mathbf{V}_{\text{tot}}^{\mathsf{T}}\mathbf{S}, && (54)\\
\mathbf{T} &= \arg\min_{\widehat{\mathbf{T}}} \left\| \widehat{\mathbf{T}}\mathbf{Q}_{\text{tot}} - \mathbf{M} \right\|_{\mathrm{F}}^{2}, & \mathbf{V}_{\text{tot}}\mathbf{T}^{\mathsf{T}} &= \mathbf{V}\mathbf{A}, && (55)\\
\mathbf{Q} &= \mathbf{A}^{-1}\mathbf{V}^{\mathsf{T}}\mathbf{S}, & \overline{\mathbf{Q}} &= \overline{\mathbf{V}}^{\mathsf{T}}\mathbf{S}, && (56)\\
\mathbf{B}_{\text{aff}} &= \arg\min_{\mathbf{B}} \left\| [\mathbf{M}^{\mathsf{T}}, \mathbf{1}]\mathbf{B} - \mathbf{Q}^{\mathsf{T}} \right\|_{\mathrm{F}}^{2}, & (\mathbf{q}^{0})^{\mathsf{T}} &= [\mu^{\mathsf{T}}, 1]\mathbf{B}_{\text{aff}}, && (57)\\
\widetilde{\mathbf{u}}(\mathbf{q}, \mu) &= \mathbf{u}_{\text{aff}}(\mu) + \mathbf{V}\mathbf{A}\mathbf{q} + \overline{\mathbf{V}}\mathcal{N}_{\theta}(\mathbf{q}), & \mathbf{W} &= \frac{\partial \widetilde{\mathbf{u}}}{\partial \mathbf{q}}, && (58)\\
\mathbf{g}_{\text{MAW}} &= \sum_{e \in \mathcal{Z}_{\text{res}}} \xi_e^{\text{res}}(\mathbf{q})\mathbf{W}_e^{\mathsf{T}}\mathbf{r}_e, & \mathbf{K}_{\text{MAW}}\Delta\mathbf{q} &= -\mathbf{g}_{\text{MAW}}, && (59)\\
\overline{\mathbf{S}}_{\text{HR}} &\approx \frac{1}{A_0} \sum_{e \in \mathcal{Z}_{\sigma}} \xi_e^{\sigma}(\mu) A_e \left\langle \mathbf{S}_e \right\rangle_e . & & && (60)
\end{aligned}
$$